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Investing in the bigger picture

International Day of Clean Energy

By UNDP Sri Lanka · January 26th, 2025

Energy lies at the core of a double challenge: leaving no one behind and protecting the planet. And, clean energy is crucial to its solution.

In a world grappling with climate change, clean energy plays a vital role in reducing emissions, and can also benefit communities lacking access to reliable power sources.

Cultivating Green Futures in the Heart of Sri Lanka

Kurunegala is one of the biggest cities in the North-Western province of Sri Lanka and one of the island's most populous districts. A 45-minute drive down the Kandy-Kurunegala Road is the modest township of Mawathagama. It's a calm and slow pace - most places on the island outside of Colombo are. The people embrace a laidback rustic way of life - simplicity is an integral part of the culture here.

The main road twists and turns through the mild rolling hills at the foot of Sri Lanka's mountainous Central Province and winds through green landscapes and coconut estates. An inconspicuous turn to the right leads to a beautiful estate lined with coconut

trees and interspersed with a wide variety of fruit bearing plants. At the end of the road is a facility maintained under the supervision of the North-Western Provincial Council.

This facility houses an office and tissue culture lab that specializes in the propagation of banana plants.

New beginnings

The Mawathagama Tissue Culture Lab has been in operation for 10 years, but its latest iteration as a private limited company only came about in the last two years. This was done to maintain the facility through the funding shortage that resulted due to the COVID-19 pandemic and the economic crisis that ensued in 2022. The facility is still not completely self-sufficient and relies on some maintenance support from the Provincial Council. However, Mr. Damith - the Agriculture Promotion Officer and the operational head at the facility - states that they expect to make enough profits to be completely self-sufficient within the next year.

The lab specialises in banana plant propagation, and they currently produce tissue-culture plants of several banana varieties including export favourites such as Ambul (Sour), Cavendish and Amban.

“Currently the demand for banana plants grown from tissue cultures is really good due its unique features, but we hope to expand into growing orchids, anthuriums and pineapple in the future”.

Building capacity

“We get orders for over 5000 plants a month - but we can only produce 750-1000 due to our current capacity and the high energy cost,” says Mr. Damith, the Agri Promotion Officer.

The facility primarily sells plants to farmers in the North-Western Province, but through publicity given from the Agri information centre in Gannoruwa, farmers from all over the island have started requesting plants.

He goes onto say, “In Rajanganaya, many farmers who grow Ambul Bananas to export come to us for plants now.”

Reducing overheads, reducing prices

The main expense and overhead of the laboratory is electricity. It makes up at least 50% of the price at which plants are sold. Through the “Enhancing Food and Energy Security through the promotion of Green Technologies & Renewable Energy for wider uptake among Vulnerable Smallholder Farmers in Sri Lanka” project, funded by the Japanese Supplementary Budget (JSB) and UNDP’s Climate Promise initiative, a 10kW Grid Tied Solar PV system was provided to the laboratory.

This solar system provides for 50% of their energy requirement and in turn reduces the cost of the plants by 25%, which directly impacts the livelihoods of farmers that rely on the high-yield, durable plants the facility provides them with. The facility makes an annual energy saving of 13,680 kWh and a financial saving of 615,000 LKR.

There are few reasons why farmers prefer plants grown from tissue cultures as opposed to the traditional method of sectioning the banana root to get multiple plants from the same mother stock. The tissue cultured plants are noticeably immune to the devastating Panama Disease and the farmers can harvest their crop at a predetermined date. In Sri Lanka, the price of fruits fluctuates seasonally, and farmers are inclined to harvest their crops during the period when the price of fruits are at their peak.

However, this cannot be planned with traditionally grown crops.

The Lab can now provide these farmers with a stock of plants that are the same age, immune to disease and grown from a high-yielding stock with unique features.

"The lab simulates natural conditions, we keep LEDs running for 16 hours a day and keep the cultures in darkness for 8 hours, which is expensive. We also need to maintain the temperature between 23°C - 26°C."

Investing in the bigger picture

By investing in a facility like this, the project indirectly supports over 50 farming families who purchase plants from the tissue culture lab. It also helps to promote the uptake of new technologies. By reducing the overhead costs of the laboratory, they can in turn reduce the price at which they sell sturdy and high-yielding stock plants to farmers across the island.

Clean Energy for All and for our Planet

The connection between clean energy, socio-economic development, and environmental sustainability is crucial in addressing issues faced by vulnerable communities worldwide. Adopting clean energy is integral to the fight against climate change, while improving energy efficiency.

This International Day for Clean Energy is a reminder on the importance of investing in people and in the bigger picture, that supports and strengthens the ecosystems in which vulnerable communities operate within. Together, we can raise awareness and mobilize action for a just and inclusive transition to clean energy for the benefit of people and the planet.

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